
MA 605: Statistical Data Analysis

Course Code	MA 605
Course Name	Statistical Data Analysis
L-T-P-C	2-1-0-3
Prerequisites	MA 202 for undergraduate
Intended for	MS/ Ph.D., Undergraduate (3rd and 4th year)
Distribution	Elective
One-time approval	10th Senate (Jun 2016)

Course Contents

- **Exploratory analysis of time series** [4 Lectures]

Introduction, examples, simple descriptive techniques, trend, seasonality, stochastic and deterministic approaches; numerical and experimental data sets; challenges in data analysis and data graphical representation, interpretation; statistical tests, significance and power of a test, choice of the critical region, constructing test statistics: the Fisher discriminant, mean and variance test, testing goodness-of-fit, chi2-test, p-values; stationary time series process (ARMA Processes).

- **Analysis of stochastic series** [12 Lectures]

Model identification and non-stationary time series models; forecasting with classical regression models; Forecasting with autocorrelations; Forecasting with lagged dependent variable; Forecast error statistics and evaluation; singularity detection, spectral density function, the periodogram, spectral analysis, correlogram, wavelet cross-correlation, multi-resolution analysis, examples and applications.

- **Clustering data techniques** [6 Lectures]

Principal component analysis; different techniques of data clustering

Text Books

1. Peck and Devore, **Statistics: The Exploration and Analysis of Data**, 7th Edition, Thomson-Brooks/Cole, 2012.
2. Montgomery, D., Jennings, C.L. and Kulahci, M., **Introduction to Time Series Analysis and Forecasting**, Hoboken, Wiley-Interscience, 2008.
3. Chatfield, C., **The Analysis of Time Series**, 6th Edition Chapman & Hall/CRC, 2004.

References

1. Petre Stoica and Randolph L. Moses, **Introduction to Spectral Analysis**, Prentice Hall, 1997.
2. Robert H. Shumway and David S. Stoffer, **Time Series Analysis and Its Applications with R Examples**, 3rd edition, Springer Texts in Statistics, 2006.
3. Raghuveer M. Rao and Ajit S. Bopardikar, **Wavelet Transform**, Pearson Education, 1998.